

Lecture 05 - Exercises

These exercises accompany Lecture 05, “Distributions and the Normal Model.” They practice the core ideas of continuous probability: density curves, interval probabilities, uniform and exponential distributions, the normal model, cumulative probabilities, z-scores, fitted distributions, and careful model checking.

The exercises are separated into two parts. The first part should be done by hand. This is important because manual work helps you understand the mechanics: what kind of distribution is being used, what interval is being measured, where the density comes from, how a z-score is built, and what a model probability means. Only after that does it become useful and convenient to move to larger datasets and use Python.

The second part uses Python. These exercises apply the same ideas to empirical return and waiting-time datasets. Python is used to summarize distributions, draw histograms, fit normal benchmarks, compute empirical frequencies, compare model probabilities with observed frequencies, and write careful interpretations.

Solutions are not included here. They will be provided separately: the by-hand solutions in Markdown and the Python solutions in a notebook.

Part A - By-Hand Practice

Exercise 1 - Discrete or Continuous?

Classify each variable as discrete or continuous. Explain briefly.

1. Number of customer complaints next week.
2. Delivery time measured in minutes.
3. Whether a borrower defaults, coded as 1 or 0.
4. Daily log return of a stock.
5. Customer satisfaction score from 1 to 5.
6. Time until the next customer arrives.
7. Monthly demand measured in units, when demand is modeled as approximately normal.

Exercise 2 - Empirical Distribution or Probability Model?

For each statement, say whether it describes an empirical distribution or a probability model. Explain briefly.

1. In a sample of 120 monthly returns, 36 were negative.
2. Under our model, the probability of a negative monthly return is 30%.
3. A histogram of observed waiting times shows a long right tail.
4. We assume daily demand follows a normal distribution with mean 100 and standard deviation 15.
5. In the observed data, 8.5% of daily returns were below -3%.
6. A fitted normal curve assigns probability 14.1% to returns below -3%.

Exercise 3 - Continuous Probabilities in Words

Delivery time T , measured in days, is treated as continuous.

For each probability, explain what it means in words.

1. $P(T \leq 2)$
2. $P(T > 5)$
3. $P(1 < T \leq 3)$
4. $P(T = 2)$

Then explain why interval and threshold probabilities are more useful than exact-value probabilities for continuous variables.

Exercise 4 - Valid Probability Density

A function $f(x)$ is proposed as a probability density on the interval $[0, 4]$:

$$f(x) = 0.25$$

for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise.

1. Is $f(x)$ ever negative?
2. Compute the total area under the curve.
3. Is this a valid probability density?
4. Compute $P(1 < X < 3)$.
5. Interpret the result in words.

Exercise 5 - Uniform Distribution

Let

$$X \sim U(10, 30).$$

1. What is the density height on the interval $[10, 30]$?
2. Compute $P(12 < X < 18)$.
3. Compute $P(X > 25)$.
4. Compute $P(X < 10)$.
5. Explain why equal-length intervals have equal probabilities.

Exercise 6 - Reading Distribution Shape

An analyst describes customer waiting times as follows:

Most customers wait less than 5 minutes, but a small number wait more than 20 minutes.
Waiting time cannot be negative.

1. Is the distribution likely symmetric or skewed?
2. If skewed, in which direction?
3. Would a normal model be a natural first choice? Explain.
4. Which distribution from the lecture is more naturally associated with waiting times?
5. Give one business question for which the right tail matters.

Exercise 7 - Normal Parameters

Suppose demand is modeled as

$$X \sim N(100, 15^2).$$

1. What is the mean?
2. What is the standard deviation?
3. Which value is more typical: 102 or 160? Explain.
4. If the standard deviation were 5 instead of 15, what would happen to the curve?
5. If the mean were 120 instead of 100, what would happen to the curve?

Exercise 8 - Z-Scores

Monthly sales growth is modeled with mean 2% and standard deviation 3%.

Compute and interpret the z-score for:

1. sales growth of 8%;
2. sales growth of -4%;
3. sales growth of 2%;
4. sales growth of 11%.

Exercise 9 - Normal Probability Below a Threshold

Suppose monthly returns are modeled as

$$R \sim N(0.01, 0.04^2).$$

Use the following standard normal probabilities:

$$P(Z < -0.25) = 0.4013, \quad P(Z < -2) = 0.0228, \quad P(Z < 1) = 0.8413.$$

Compute and interpret:

1. $P(R < 0)$;
2. $P(R < -0.07)$;
3. $P(R < 0.05)$.

Be explicit that these are probabilities under the normal model.

Exercise 10 - Normal Probability in an Interval

Suppose

$$X \sim N(100, 15^2).$$

Use:

$$P(Z < 1.33) = 0.9082, \quad P(Z < -0.67) = 0.2514.$$

1. Convert $X = 120$ to a z-score.
2. Convert $X = 90$ to a z-score.
3. Compute $P(90 < X < 120)$.
4. Interpret the result.

Exercise 11 - The 68-95-99.7 Rule

Assume a variable is normally distributed with mean 100 and standard deviation 15.

Use the empirical rule to answer:

1. Approximately what percentage of values lie between 85 and 115?
2. Approximately what percentage lie between 70 and 130?
3. Approximately what percentage lie between 55 and 145?
4. What assumption is needed to use these percentages?
5. Why should we be careful when applying this rule to real data?

Exercise 12 - Comparing Empirical and Model Probabilities

For a fitted normal model of daily log returns, suppose the following comparison is obtained:

Event	Empirical frequency	Normal model probability
$-0.02 < X < 0.02$	68.0%	51.0%
$X < \mu - 3\sigma$	0.70%	0.135%

1. For the first event, does the normal model put more or less probability than the data?
2. For the second event, does the normal model put more or less probability than the data?
3. Why is the second row especially important in finance?
4. Write one careful sentence interpreting the comparison.

Exercise 13 - Normal Model: Useful but Dangerous

For each statement, say whether it is careful or too strong. Rewrite the statements that are too strong.

1. "The histogram is roughly bell-shaped, so a normal model may be a useful benchmark."
2. "The histogram is roughly bell-shaped, so the data are definitely normal."
3. "Under the normal model, the probability of a return below -7% is 2.28%."
4. "A z-score of -3 means the outcome cannot happen."
5. "Financial returns may have heavier tails than a normal distribution."
6. "The normal model is useless because it is never exactly true."

Exercise 14 - Choosing a Distribution

For each situation, choose the most plausible first model among uniform, exponential, and normal. Explain briefly.

1. A random number generator produces any value between 0 and 1 with equal density.
2. Time until the next support call arrives.
3. Measurement error from a precise instrument centered near zero.
4. Daily demand for a mature product, fluctuating around a stable mean.
5. Waiting time at a shop, where most customers wait briefly but a few wait a long time.

Part B - Python Applications**Exercise 15 - Empirical Distribution of Monthly Returns**

Use the file:

`data and notebooks/Lecture 05 monthly returns.csv`

The relevant column is `monthly_return`.

1. Load the dataset.
2. Convert the date column to dates.
3. Compute the sample size, mean, median, standard deviation, minimum, maximum, and quartiles.
4. Plot a histogram of monthly returns.
5. Estimate the empirical probability that the monthly return lies between 1% and 2%.
6. Write a short interpretation of the distribution shape.

Exercise 16 - Waiting Times and Skewness

Use the file:

`data and notebooks/Lecture 05 shop waiting times.csv`

The relevant column is `waiting_time_minutes`.

1. Load the dataset.
2. Compute the mean, median, standard deviation, minimum, maximum, and selected percentiles.
3. Plot a histogram of waiting times.
4. Estimate the empirical probability that waiting time is between 1 and 2 minutes.
5. Compare the mean and median.
6. Write a short interpretation of why the normal model may be questionable for waiting times.

Exercise 17 - Uniform Distribution in Python

Let $X \sim U(10, 30)$.

1. Create a small Python function that computes $P(a < X < b)$ for this uniform distribution.
2. Use it to compute $P(12 < X < 18)$, $P(25 < X < 30)$, and $P(5 < X < 12)$.
3. Explain how the function handles intervals partly outside $[10, 30]$.

Exercise 18 - Normal Demand Probability

Assume daily demand follows

$$X \sim N(100, 15^2).$$

1. Compute $P(90 < X < 120)$ using a normal CDF.
2. Compute the z-scores for 90 and 120.
3. Recompute the probability using the standard normal CDF and the z-scores.
4. Confirm that the direct and standardized calculations agree.

Exercise 19 - Fitting a Normal Model to Netflix Returns

Use the file:

`data and notebooks/Lecture 05 nflx yahoo daily log returns 2015 2026.csv`

The relevant column is `log_return`.

1. Load the dataset.
2. Compute the sample mean and sample standard deviation of daily log returns.
3. Plot a histogram of log returns.
4. Overlay or describe the fitted normal benchmark.
5. Report the fitted values of $\hat{\mu}$ and $\hat{\sigma}$.

Exercise 20 - Empirical Versus Normal Probabilities

Using the Netflix log returns from Exercise 19:

1. Estimate the empirical probability of $-0.02 < X < 0.02$.
2. Estimate the empirical probability of $-0.05 < X < -0.02$.
3. Estimate the empirical probability of $X < -0.03$.
4. Estimate the empirical probability of $X > 0.04$.
5. Compute the same probabilities under the fitted normal model.

6. Put the empirical and model probabilities in one table.
7. Write a short interpretation of where the model overestimates or underestimates probabilities.

Exercise 21 - Tail Risk Under the Normal Model

Using the Netflix log returns from Exercise 19:

1. Compute the threshold $\hat{\mu} - 3\hat{\sigma}$.
2. Estimate the empirical probability that $X < \hat{\mu} - 3\hat{\sigma}$.
3. Compute the normal-model probability that $X < \hat{\mu} - 3\hat{\sigma}$.
4. Compare the empirical and model probabilities.
5. Explain why this comparison matters for risk analysis.

Exercise 22 - Careful Probability Memo

Write a short memo, using the results from Exercises 19-21, that includes:

1. the fitted mean and standard deviation;
2. one empirical probability;
3. the corresponding normal-model probability;
4. one sentence explaining the difference between empirical frequencies and model probabilities;
5. one caution about using the normal model for financial returns.